

US00PP36346P2

(12)

United States Plant Patent

Rosas et al.

(10) Patent No.:

US PP36,346 P2

(45) Date of Patent:

Dec. 31, 2024

(54) STRAWBERRY PLANT VARIETY NAMED  
'Plared 18145'

(50) Latin Name: *Fragaria x ananassa*  
Varietal Denomination: Plared 18145

(71) Applicant: Plantas de Navarra S.A., Valtierra  
(ES)

(72) Inventors: Manuel Rosas, Anderson, CA (US);  
Pedro Dominguez, Anderson, CA (US)

(73) Assignee: Plantas de Navarra S.A., Valtierra  
(ES)

(\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 0 days.

(21) Appl. No.: 18/445,832

(22) Filed: Feb. 23, 2024

(51) Int. Cl.  
A01H 5/08 (2018.01)  
A01H 6/74 (2018.01)

(52) U.S. Cl.  
USPC ..... Plt./208  
CPC ..... A01H 6/7409 (2018.05)

(58) Field of Classification Search  
USPC ..... Plt./208  
CPC ..... A01H 6/7409; A01H 5/08  
See application file for complete search history.

(56) References Cited

U.S. PATENT DOCUMENTS

PP20,552 P3 12/2009 Shaw et al.  
PP31,259 P3 12/2019 Pierron-Darbonne

Primary Examiner — Keith O. Robinson

(74) Attorney, Agent, or Firm — Hunt IP Law

(57) ABSTRACT

The new and distinct variety of strawberry plant variety  
'Plared 18145' is provided. The variety can be distinguished  
by its outstanding features of an upright plant architecture  
and darker red fruit and firmer long conical fruit.

8 Drawing Sheets

1

Latin name of the genus and species:  
Botanical classification:  
a. Genus—*Fragaria*.  
b. Species—*x ananassa*.  
Variety denomination: The new strawberry plant claimed  
is of the variety denominated 'Plared 18145'.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a new and distinct annual  
variety of strawberry plant, which has been given the variety  
denomination of 'Plared 18145'.

Background of the Related Art

Cultivated strawberry is a hybrid species of the genus  
*Fragaria* that is grown worldwide for its fruit. Modern  
strawberry was first bred in Brittany, France, in the 18<sup>th</sup>  
century by crossing *Fragaria virginiana* with *Fragaria*  
*chiloensis*. Strawberry fruit is an aggregate accessory fruit,  
with the fleshy part of the fruit being derived from the  
receptacle that holds the ovaries.

Strawberry varieties vary widely in color, size, shape,  
flavor, season of ripening, degree of fertility, and suscepti-  
bility to disease. Certain varieties vary in foliage, and some  
vary in the relative development of their reproductive  
organs. Typically, strawberry flowers appear hermaphroditic  
in structure, but function as either male or female. Generally,  
commercial production of strawberry plants involves propa-  
gation from runners and distribution as either plugs or bare  
root plants. Cultivation is either perennial or annual plasti-  
culture. During the off season, strawberries can also be  
produced in greenhouses.

2

Strawberry fruit is widely appreciated for its characteristic  
bright red color, aroma, juicy texture, and sweetness. Straw-  
berry fruit is a popular fruit that is generally consumed either  
fresh or in prepared foods, such as preserves and baked  
goods.

Strawberries are an important and valuable fruit crop.  
Accordingly, there is a need for new varieties of strawberry  
plants. In particular, there is a need for improved varieties of  
strawberry plant that are stable, high yielding, and agro-  
nominically sound.

SUMMARY OF THE INVENTION

The present invention relates to a new and distinct annual  
variety of strawberry, which has been given the variety  
denomination of 'Plared 18145'. Its market class is that of  
summer planted variety. 'Plared 18145' is intended for use  
as a frigo plant to be planted in the summer and produce  
fresh fruit in the fall and beyond.

The new strawberry variety is a selection resulting from  
a sexual cross of strawberry plants at Watsonville, California  
in 2018, involving a seed parent known as 'Plared 0822'  
(U.S. Plant Pat. No. 31,259) and a pollen parent known as  
'14-01' (unpatented).

The selection was subsequently evaluated for four years  
in Watsonville, California.

Asexual reproduction of the new variety by cutting propa-  
gation since 2018 in Watsonville, California has demon-  
strated that the new variety reproduces true to type with all  
of the morphological characteristics, as herein described,  
firmly fixed and retained through successive generations of  
such asexual propagation.

Selection criteria was based on the needs the California  
fresh fruit market and was optimized for the summer plant-  
ing for targeted production ideotype.



The following characteristics of the new variety have been repeatedly observed and can be used to distinguish ‘Plared 18145’ as a new and distinct variety: upright plant architecture, darker red fruit, firmer long conical fruit.

Compared to the seed parent ‘Plared 0822’, plants of the new variety present more vigor than ‘Plared 0822’. In terms of the fruit shape, ‘Plared 18145’ presents an elongated conical shape and ‘Plared 0822’ presents a short conical shape. Compared to the pollen parent ‘14-01R’, the fruits of ‘14-01R’ are darker than ‘Plared 18145’. In terms of architecture, ‘14-01R’ is an open variety and ‘Plared 18145’ presents very compact architecture.

Plants of the new variety differ from the known reference variety ‘Portola’ (U.S. Plant Pat. No. 20,552) as shown in Table 1.

TABLE 1

| Comparison of ‘Plared 18145’ and ‘Portola’ |                   |                  |
|--------------------------------------------|-------------------|------------------|
| Characteristic                             | ‘Plared 18145’    | ‘Portola’        |
| Density of Foliage                         | Medium            | Strong           |
| Fruit Shape                                | Elongated Conical | Conical          |
| Fruit Color                                | sRGB 168, 0, 23   | sRGB 246, 71, 65 |

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying photographic illustrations show the typical appearance of the new variety ‘Plared 18145’. The colors are as nearly true as is reasonably possible in a color representation of this type. Colors in the photographs may differ slightly from the color values cited in the detailed botanical description which accurately describes the colors of the new plant.

FIG. 1 is a photograph of the new variety ‘Plared 18145’, 100 days after planting, demonstrating its medium red ovate to oblong fruit.

FIG. 2 is a photograph of the new variety ‘Plared 18145’, demonstrating its upright architecture and summer planting adaptation. This photo was taken on Sep. 7, 2022, 82 days after planting using frigo plants.

FIG. 3 is a line graph of marketable yield over three summer planting seasons of new variety ‘Plared 18145’ and industry reference ‘Portola’.

FIG. 4 is a photograph of the new variety of ‘Plared 18145’, 100 days after planting.

FIG. 5 is a photograph of the new variety of ‘Plared 18145’, 100 days after planting.

FIG. 6 is a photograph of the new variety of ‘Plared 18145’, 100 days after planting.

FIG. 7 is a photograph of the new variety of ‘Plared 18145’, 100 days after planting.

FIG. 8 is a photograph of the new variety of ‘Plared 18145’, 100 days after planting.

DETAILED BOTANICAL DESCRIPTION

The following detailed description sets forth the distinctive characteristics of ‘Plared 18145’. The datum which defines these characteristics was collected from asexual reproductions of the original selection. Dimensions, sizes, colors, and other characteristics are approximations and averages set forth as accurately as possible. The plant history was taken on plants approximately 175 days old, and the

descriptions relate to plants grown in Watsonville, California. Color notations are in reference to the standard RGB (sRGB) color space.

Classification:

- a. *Family*.—Rosaceae.
- b. *Botanical*.—*Fragaria*.
- c. *Common name*.—Strawberry.
- d. *Variety name*.—‘Plared 18145’.

Parentage:

- a. *Female parent*.—‘Plared 0822’ (U.S. Plant Pat. No. 31,259).
- b. *Male parent*.—‘14-01R’. (unpatented).

PLANT

General:

- a. *Height*.—25 cm.
- b. *Diameter*.—30 cm.
- c. *Number of crowns per plant*.—6.
- d. *Growth habit*.—Upright.
- e. *Density of foliage*.—Medium to High Density.
- f. *Vigor*.—Medium.
- g. *Stolon*.—i. Average number of daughter plants per sq. ft. — 4. ii. Diameter at bract — 0.95 cm. iii. Anthocyanin color — Medium Red (sRGB 250, 60, 75). iv. Stolon color — Medium Green (sRGB 0, 73, 5). v. Density of pubescence on the stolon — Medium.
- h. *Hardiness zone*.—16° C.-27° C. Hardiness zone 13.

LEAVES

General:

- a. *Number of leaflets*.—3.
- b. *Color of upper surface*.—Dark Green (sRGB 47, 68, 0).
- c. *Color of lower surface*.—Light Green (sRGB 105, 122, 131).
- d. *Leaf blistering*.—Strong.
- e. *Leaf glossiness*.—Strong.
- f. *Variegation*.—Absent.
- g. *Terminal leaflet*.—i. Length — 6.1 cm. ii. Width — 5.9 cm. iii. Length/width ratio — 1. iv. Number of teeth/terminal leaflet — 23. v. Shape of base — Rounded. vi. Margin — Serrate to Crenate. vii. Shape in cross section — Concave to Straight. viii. Shape of apex — Rounded.
- a. *Petiole*.—i. Length — 17 cm. ii. Diameter — 0.5 cm. iii. Attitude of hairs — Upwards and Outward. iv. Bract frequency (number present on each petiole) — N/A. v. Color — Light Green (sRGB 179, 232, 144).
- b. *Petiolule*.—i. Length — Long. 10-12 mm. ii. Color — Light Green (SRGB 161, 226, 160).
- c. *Stipule*.—i. Length — 2.9 cm. ii. Width — 1.1 cm. i. Anthocyanin coloration — Absent. ix. Anthocyanin color — Absent. x. Stipule color — Translucent Green (sRGB 163, 211, 174).
- d. *Rugosity*.—Medium.

INFLORESCENCE

General:

- a. *Position in relation to foliage*.—Level.
- b. *Pedicel*.—i. Attitude of hairs — Upwards. ii. Length — 20 cm. iii. Diameter — 0.3 cm. iv. Color — Light Green (sRGB 179, 232, 144).



- c. *Flower*.—i. Arrangement of petals — Overlapping.  
 ii. Stamen — Present. iii. Anther color (if stamen is  
 present) — Dark Yellow (sRGB 205, 169, 95). iv.  
 Number of flowers per inflorescence — 2-6.v. Mean  
 diameter of individual flower — 25 mm. 5  
 d. *Petal*.—i. Length/width ratio — Equal. ii. Color of  
 upper side — White (sRGB 255, 255, 255). iii. Color  
 of lower side — White (sRGB 255, 255, 255).  
 e. *Calyx*.—i. Color — Green (sRGB 0, 51, 0). ii. Mean  
 Diameter — Mean: 22 mm. 10  
 f. *Sepal*.—i. Color of surface — Dark green (sRGB 56,  
 80, 30). ii. Color of lower surface — Medium Green  
 (sRGB 141, 171, 6). iii. Attitude of sepals — Out-  
 ward to away from berry. 15

## FRUIT

## General:

- a. *Fruit weight*.—20.2 grams (2022 season average). 20  
 b. *Shape*.—Elongated conical.  
 c. *Differences in shape between primary and secondary  
 fruit*.—Moderate.  
 d. *Glossiness*.—High.  
 e. *Firmness*.—Strong Firmness. 25  
 f. *Color*.—Dark Red (sRGB 168, 0, 23).  
 g. *Evenness of fruit color*.—Even.  
 h. *Position of achenes*.—Level with Surface.  
 i. *Number of fruit per truss*.—2-4.  
 j. *Position of calyx attachment*.—Inserted. 30

- k. *Color of flesh (excluding core)*.—Dark Red (sRGB  
 168, 0, 23).  
 l. *Color of core*.—Medium red (sRGB 247, 52, 67).  
 m. *Sugar content (as soluble solids in degrees brix)*.—  
 8.6% (2022 season average).  
 n. *Production*.—i. Flowering interval — Beginning  
 approximately 2 weeks of planting in summer with  
 frigo plants, continuous flowering thereafter. ii. Har-  
 vest interval — Beginning approximately 6 weeks  
 after planting in summer with frigo plants, continu-  
 ous harvest thereafter. iii. Type of bearing — Day-  
 Neutral. iv. Productivity — Three-year average of  
 850 grams per plant produced within five months of  
 summer planting using frigo plants (e.g., planting in  
 early June and harvesting through October).  
 o. *Width of band without achenes*.—Absent or very  
 narrow.  
 p. *Adherence to calyx*.—Strong.  
 q. *Cavity size*.—Absent.  
 r. *Storage life*.—Very good storage life (10 days 4° C.),  
 better than reference Portolas (7days 4° C.).  
 s. *Market use*.—Commercial fresh market in summer  
 and fall production.  
 t. *Length*.—4.8/5.2 cm.  
 u. *Width*.—4.4/4.8 cm.  
 v. *Flowering runners*.—Present.  
 The invention claimed is:  
 1. A new and distinct variety of strawberry plant named  
 ‘Plared 18145’, as illustrated and described herein.

\* \* \* \* \*





FIG. 1





FIG. 2



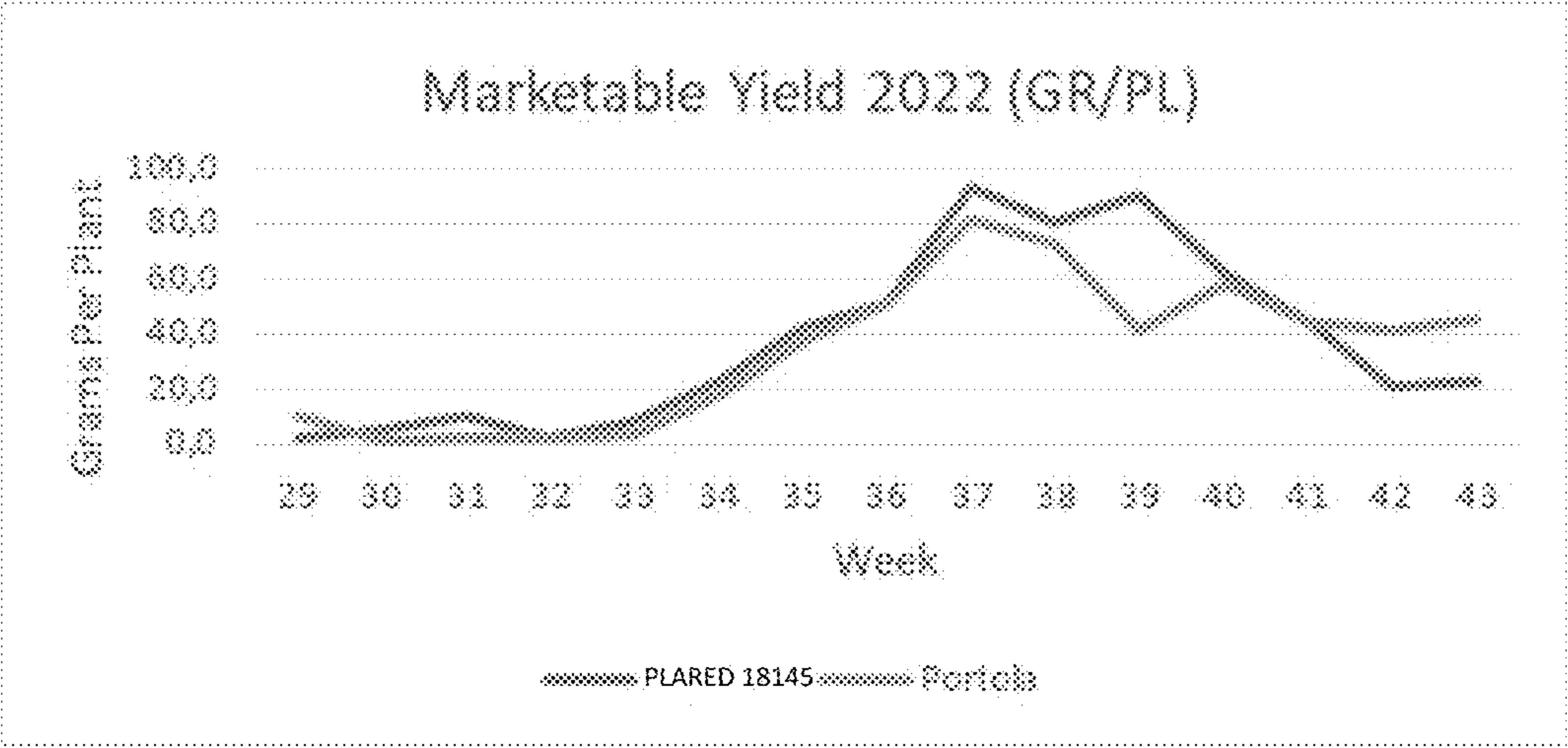


FIG. 3





FIG. 4





FIG. 5





FIG. 6





FIG. 7



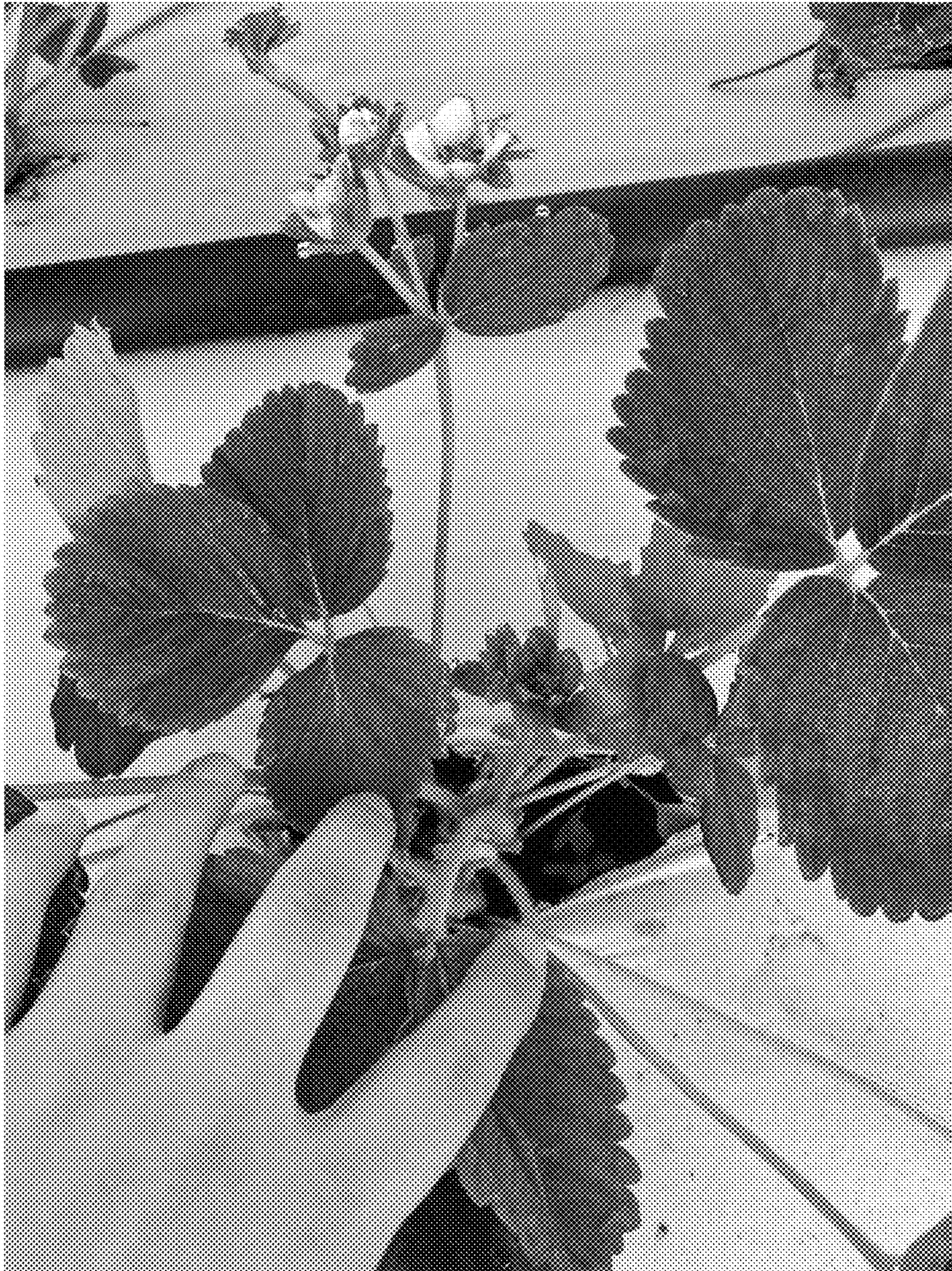


FIG. 8